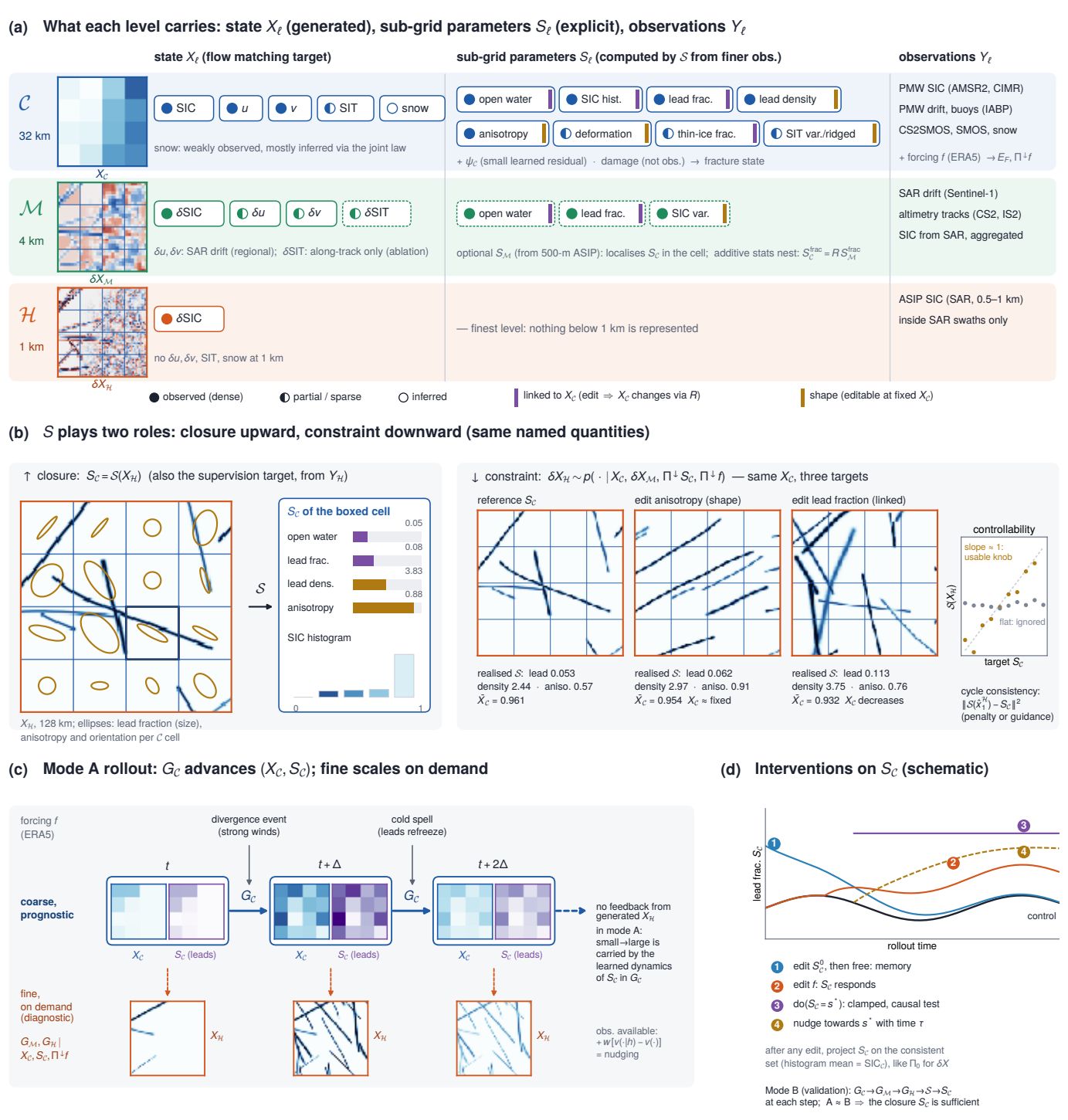

(a) What each level carries: state X_ℓ (generated), sub-grid parameters S_ℓ (explicit), observations Y_ℓ
state X_ℓ (flow matching target)
sub-grid parameters S_ℓ (computed by $\mathcal{S}$ from finer obs.)
observations Y_ℓ
$\mathcal{C}$
32 km
$X_\mathcal{C}$
SIC
u
v
SIT
snow
snow: weakly observed, mostly inferred via the joint law
open water
SIC hist.
lead frac.
lead density
anisotropy
deformation
thin-ice frac.
SIT var./ridged
+ $\psi_\mathcal{C}$ (small learned residual) · damage (not obs.) → fracture state
PMW SIC (AMSR2, CIMR)
PMW drift, buoys (IABP)
CS2SMOS, SMOS, snow
+ forcing f (ERA5) → E_F, $\Pi^{\downarrow} f$
$\mathcal{M}$
4 km
$\delta X_\mathcal{M}$
δSIC
δu
δv
δSIT
$\delta u, \delta v$: SAR drift (regional); δSIT: along-track only (ablation)
open water
lead frac.
SIC var.
optional $S_\mathcal{M}$ (from 500-m ASIP): localises $S_\mathcal{C}$ in the cell; additive stats nest: $S_\mathcal{C}^{\text{frac}} = R\, S_\mathcal{M}^{\text{frac}}$
SAR drift (Sentinel-1)
altimetry tracks (CS2, IS2)
SIC from SAR, aggregated
$\mathcal{H}$
1 km
$\delta X_\mathcal{H}$
δSIC
no $\delta u, \delta v$, SIT, snow at 1 km
— finest level: nothing below 1 km is represented
ASIP SIC (SAR, 0.5–1 km)
inside SAR swaths only
observed (dense)
partial / sparse
inferred
linked to $X_\mathcal{C}$ (edit ⇒ $X_\mathcal{C}$ changes via R)
shape (editable at fixed $X_\mathcal{C}$)
(b) S plays two roles: closure upward, constraint downward (same named quantities)
↑ closure: $S_\mathcal{C} = \mathcal{S}(X_\mathcal{H})$ (also the supervision target, from $Y_\mathcal{H}$)
$\mathcal{S}$
$S_\mathcal{C}$ of the boxed cell
open water
0.05
lead frac.
0.08
lead dens.
3.83
anisotropy
0.88
SIC histogram
0
1
$X_\mathcal{H}$, 128 km; ellipses: lead fraction (size), anisotropy and orientation per $\mathcal{C}$ cell
↓ constraint: $\delta X_\mathcal{H} \sim p(\cdot \mid X_\mathcal{C}, \delta X_\mathcal{M}, \Pi^{\downarrow} S_\mathcal{C}, \Pi^{\downarrow} f)$ — same $X_\mathcal{C}$, three targets
reference $S_\mathcal{C}$
edit anisotropy (shape)
edit lead fraction (linked)
realised $\mathcal{S}$: lead 0.053
density 2.44 · aniso. 0.57
$\bar{X}_\mathcal{C} = 0.961$
realised $\mathcal{S}$: lead 0.062
density 2.97 · aniso. 0.91
$\bar{X}_\mathcal{C} = 0.954$ $X_\mathcal{C} \approx$ fixed
realised $\mathcal{S}$: lead 0.113
density 3.75 · aniso. 0.76
$\bar{X}_\mathcal{C} = 0.932$ $X_\mathcal{C}$ decreases
controllability
slope ≈ 1: usable knob
flat: ignored
$\mathcal{S}(X_\mathcal{H})$
target $S_\mathcal{C}$
cycle consistency: $\|\mathcal{S}(\hat{X}_1^{\mathcal{H}}) - S_\mathcal{C}\|^2$ (penalty or guidance)
(c) Mode A rollout: $G_\mathcal{C}$ advances $(X_\mathcal{C}, S_\mathcal{C})$; fine scales on demand
forcing f (ERA5)
divergence event (strong winds)
cold spell (leads refreeze)
t
$t + \Delta$
$t + 2\Delta$
$G_\mathcal{C}$
$G_\mathcal{C}$
coarse, prognostic
$X_\mathcal{C}$
$S_\mathcal{C}$ (leads)
$X_\mathcal{C}$
$S_\mathcal{C}$ (leads)
$X_\mathcal{C}$
$S_\mathcal{C}$ (leads)
no feedback from generated $X_\mathcal{H}$
in mode A: small→large is carried by the learned dynamics of $S_\mathcal{C}$ in $G_\mathcal{C}$
fine, on demand (diagnostic)
$G_\mathcal{M}, G_\mathcal{H} \mid X_\mathcal{C}, S_\mathcal{C}, \Pi^{\downarrow} f$
$X_\mathcal{H}$
$X_\mathcal{H}$
$X_\mathcal{H}$
obs. available: + $w[v(\cdot|h) - v(\cdot)]$ = nudging
(d) Interventions on $S_\mathcal{C}$ (schematic)
lead frac. $S_\mathcal{C}$
rollout time
control
1 edit $S_\mathcal{C}^0$, then free: memory
2 edit f: $S_\mathcal{C}$ responds
3 do($S_\mathcal{C} = s^\star$): clamped, causal test
4 nudge towards $s^\star$ with time τ
after any edit, project $S_\mathcal{C}$ on the consistent set (histogram mean = SIC$_\mathcal{C}$), like Π_0 for δX
Mode B (validation): $G_\mathcal{C} \to G_\mathcal{M} \to G_\mathcal{H} \to \mathcal{S} \to S_\mathcal{C}$ at each step; A ≈ B ⇒ the closure $S_\mathcal{C}$ is sufficient